21 Classification of Proteins (Advanced)

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How they are different?

SOY

WHEY

CASEIN

Soy Protein

KEY FACT 1:
Good source of plant-based protein which is highly digestible (PDCAAS=1.0)

KEY FACT 2:
Complete protein that provides all essential amino acids

KEY FACT 3:
Lactose free and cholesterol-free

KEY FACT 4:
Can be taken at anytime of the day including pre and

Whey

- Quick digestion and absorption into bloodstream
- Dissipates 3 hours after ingestion

- Muscle growth promoting
- Contains more leucine

- Post-workout to supply damaged muscle tissues with protein and build muscle

Casein

- Slow digestion and absorption into bloodstream
- Dissipates 8 hours after ingestion

- Muscle preserving
- Slows down muscle protein breakdown

- Before bedtime
- Between spaced-out meals or fasting period
- Hours before workout

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Classification of Proteins (Advanced)

1. It is essential to consume sufficient protein based on our body weight and activity level.

2. So, do you remember what is your daily protein requirement?

3. Protein consumption is essential in helping you to manage your weight and body composition as well as enhancing muscle growth, post exercise recovery and immune function.

4. There are 3 main supplementary protein powders comprising soy, whey and casein.

5. Soy protein is a highly-digestible complete plant-based protein derived from soybeans where as casein as well as whey protein are animal-based protein derived from cow' s milk. However, these 3 types of proteins contain all the essential amino acids needed by our bodies.

6. Soy protein is lactose-free and cholesterol-free. It is suitable to be consumed at any time of the day including pre and post-workout sessions.

7. So, what is the main difference between whey and casein protein?

8. They differ in terms of digestion rate. Whey protein digests quickly, hence it is being absorbed faster into the bloodstream and promotes quick muscle protein synthesis. However, it stays in the body for a shorter period of time in which it begins to dissipate about 3 hours after ingestion.

9. As for casein protein, it is a slow-digesting protein providing sustained release of amino acids into the bloodstream. Thus, it stays in the body longer as it only starts to dissipate about 8 hours after ingestion.

10. Whey protein has higher leucine content that signals our body for muscle protein synthesis. Casein protein reduces muscle protein breakdown, hence it has muscle-preserving properties.

11. It is best to consume whey protein within 30 minutes to 1 hour after exercise with at least 20g of protein to get maximum new protein into your muscles. As for casein protein, it is suitable to be consumed before bedtime, between spaced out meals, during fasting period or hours before your workout session.

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22 Balance of Omega 3 & 6

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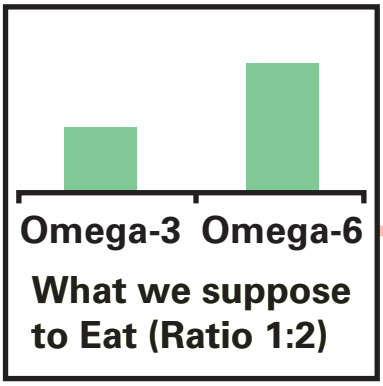

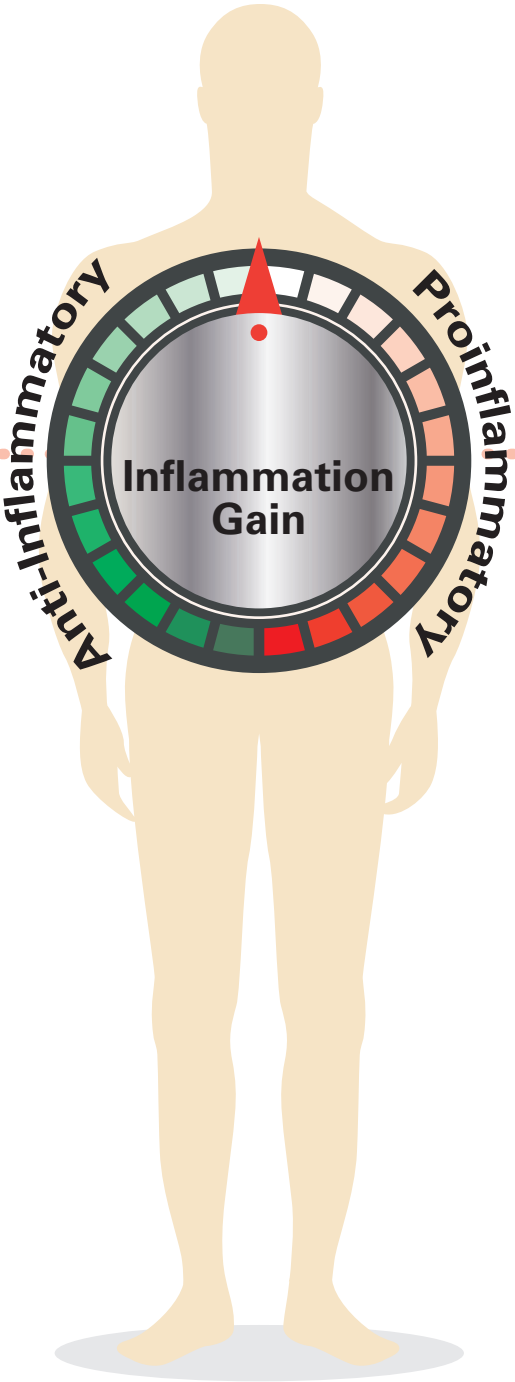

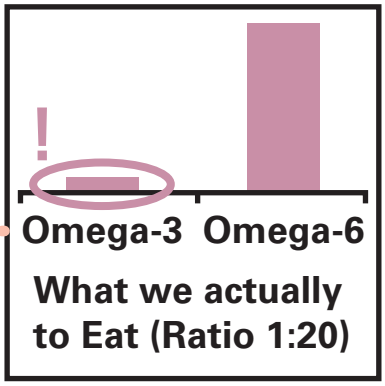

Salmon

Walnuts

Omega 3 Eggs

Flax Seed Oils

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Balance of Omega 3 & 6

Both Omega 3 & 6 are essential fatty acids which our body can’ t make them, we need to take them from our diet daily. The problems of modern day are we consume too much omega 6 and too few omega 3 in our diet.

The effect of over omega 6 consumed, it is recommended that we consume these essential fatty acids in the ratio of 2:1, which means for the 20mg of omega 6 you consume, you must consume 10mg omega 3 to counter the effects.

These fatty acids are found in meats, margarines, peanuts, and many of our processed foods. The major sources of omega 3 fatty acids are found in cold water fish, flaxseed, soybeans, organic eggs, sardines, salmon, flax seeds and walnuts.

Few ways to reduce your omega-6 intake:

1. Change your cooking oils from vegetable oils to omega-3 rich oils
2. Limit processed foods, deep fried, fast food, etc
3. Reduce dressings, margarine, and mayonnaises
4. Consume low fat or fat free foods

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23 Tips and Tricks on Food Label

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Tips 1:
Choose low in Saturated fat, Added Sugar and Sodium

Tips 2:
Get enough Vitamins, Minerals and Fibre

Nutrition Information		
Servings per package - 16 Serving size - 30g(2/3 cup)		
	Per serve	**Per 100g**
Energy	**432kj**	**1441kj**
Protein	2.8g	9.3g
Fat		
Total	0.4g	1.2g
Saturated	0.1g	0.3g
Carbohydrate		
Total	18.9g	62.9g
Sugars	3.5g	11.8g
Fibre	6.4g	21.2g
Sodium	65mg	215mg
Ingredients: Cereals(76%)(wheat, oatbran, barley), psyllium husk(11%), sugar, rice, malt extract, honey, salt, vitamins		

Total Fat :
Generally choose foods with less than **10g per 100g.**
For milk, yogurt, ice cream, choose less than **2g per 100g.**
For cheese,choose less than **15g per 100g.**
Saturated Fat :
Aim for the lowest, per 100g.
Less than 3g per 100g is best.

Keyword for high saturated fat:
animal fat/oil, butter, milk solid, coconut oil/ milk/cream, sour cream, vegetable shortening

Sugars:
Avoid sugar completely is not necessary but try to avoid larger amounts of added sugars.if sugar content per 100g is more than 15g, check that sugar (or alternative names for added sugar) is not listed high on the ingredient list.

Keyword for added sugars:
dextrose, fructose, glucose, any syrup, malt, maltose, cane sugar, raw sugar, sucrose

Ingredient :
Listed from greatest to smallest by weight. Use this to check the first three ingredient for items high in saturated fat, sodium (salt) or added sugar.

Sodium (Salt)
Choose lower sodium options among similiar foods. **Food with less than 400mg per 100g are good , and less than 120mg per 100g is best.**

Keyword for salt :
baking powder, salt, soda, sodium, meat/yeast extract , monosodium glutamate (MSG), stock cubes

Fibre :
Not all labels include fibre. Choose breads and cereals with **3g or more per serve**

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Tips and Tricks on Food Label

1. **Step 1: Choose food that are low in Saturated fat, Added Sugar and Sodium**
   • Fats can be categorised into:
   i. Total - Choose food with less than 10g per 100g
   ii. Saturated - It is linked to increase risk of heart disease and high blood cholesterol. Limit to 3g per 100g.
   Keyword for ingredient high in saturated fat: animal fat/oil, butter, milk solid, coconut oil/milk/cream, sour cream, vegetable shortening.
   iii. Trans - Created in food processing that can increase risk of developing heart disease and stroke. Possibly replace trans fat with monounsaturated or polyunsaturated fats.

   • Sugar is a type of carbohydrate. Total sugars includes the sugars naturally present and sugars added during processing of food. If sugar content is more than 15g per 100g, check that sugar is not listed high on the ingredient list.
   Keyword for added sugars: dextrose, fructose, glucose, any syrup, malt, maltose, raw sugar, sucrose

   • Sodium indicates how much salt the food contains. Lower sodium options are always the choice. Food with less than 400mg per 100g are good, less than 120mg per 100g is best.
   Keyword for salt: baking powder, salt, soda, sodium, meat/yeast extract, monosodium glutamate(MSG), stock cubes

2. **Step 2: Get enough Vitamins, Minerals and Fibre**
   • Eat more potassium, vitamin D, calcium and iron to reduce risk of certain health problems such as osteoporosis.
   • Not all labels include fibre. Choose breads and cereals with 3g or more per serving.
   • Choose fresh fruits and vegetables to get more of these nutrients

   Foods with more than one ingredient must have an ingredient list on the label. The ingredients are listed from greatest to smallest by weight. Those in the largest amount are listed first. This is helpful to individuals with food sensitivities, vegetarians or people who wish to limit added sugars.

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24 Exercise

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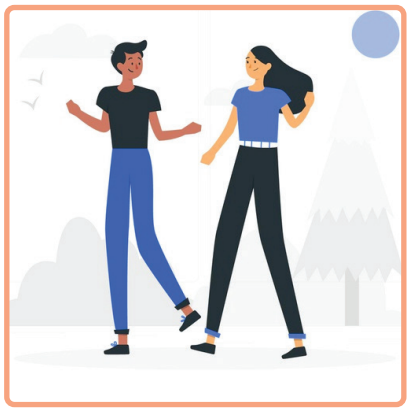
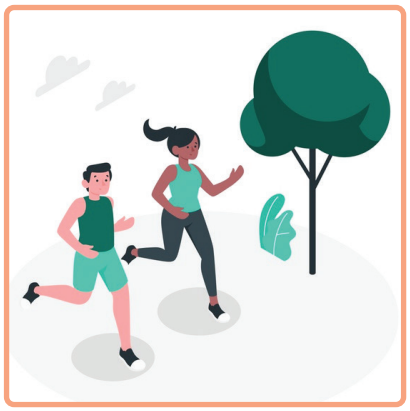
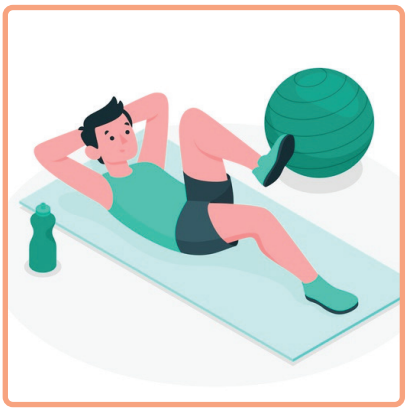
80% Nutrition 20% Exercise 100% Mindset



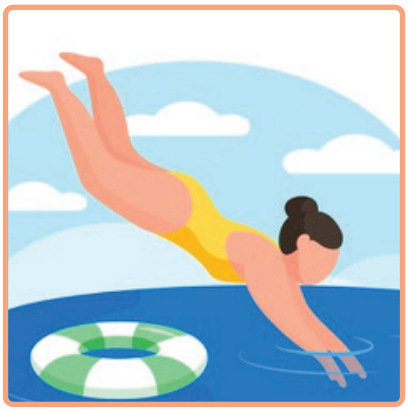
GETTING STARTED



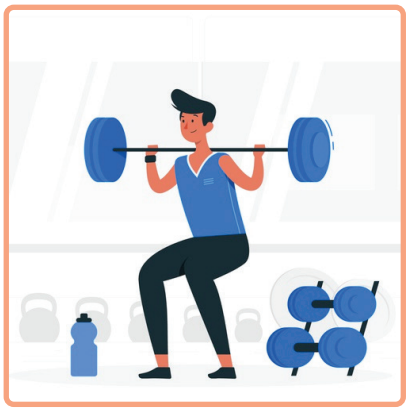
MODERATE -INTENSITY AEROBIC ACTIVITIES



VIGOROUS -INTENSITY AEROBIC ACTIVITIES



STRENGTH TRAINING (BUILD MUSCLES, BURN FATS)



HYPERTROPHY TRAINING

Preparation:
solid meal best taken 30-60minutes before exercise;
liquid meal best taken 30minutes before exercise.

Recovery:
Protein-based meal within 30minute after exercise.

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Exercise

1. (Customer' s name) what kind of exercise you usually do?

2. Adults aged 18–64 should do at least 150 minutes of moderate-intensity aerobic physical activity throughout the week.

3. We should perform activities in multiple shorter bouts, of at least 10 minutes each, spread throughout the week then adding together the time spent during each of these bouts: e.g. 30 minutes of moderate-intensity activity 5 times per week.

Getting started	Moderate intensity aerobic activities	Vigorous intensity aerobic activities	Strength training (build muscles, burn fats)	Hypertrophy training
- to start walking brisk walk for five or ten minutes a few times a day	- brisk walking (at least 2.5 miles per hour) - water aerobics - dancing (ballroom or social) - gardening - tennis (doubles) - biking slower than 16km (10 miles) per hour	- hiking uphill or with a heavy backpack - running - swimming laps - aerobic dancing - heavy yard work like continuous digging or hoeing - tennis (singles) - cycling 10 miles per hour or faster	-Start with two days for two to three weeks, then add a third day. -Squats -Push ups (on an incline if needed) -Inverted rows -Planks -Most training can be done seated for those with mobility and balance issues.	Use a reps and rest cycle. Weightlifters should aim for 6-12 reps per set. Allow 60-90 seconds between sets for rest. This will help achieve hypertrophy because your muscles will be fatigued.

4. Despite regular exercise, we should take care of pre-workout and post-workout meal to maximise the health benefit from activities.

a. Preparation: solid meal (consists of carbohydrates and protein) best taken 30-60minutes before exercise; liquid meal best taken 30minutes before exercise.

b. Recovery: Protein-based meal (consists of whey, branched chain amino acid, L-glutamine) within 30minute after exercise.

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25 Digestive System

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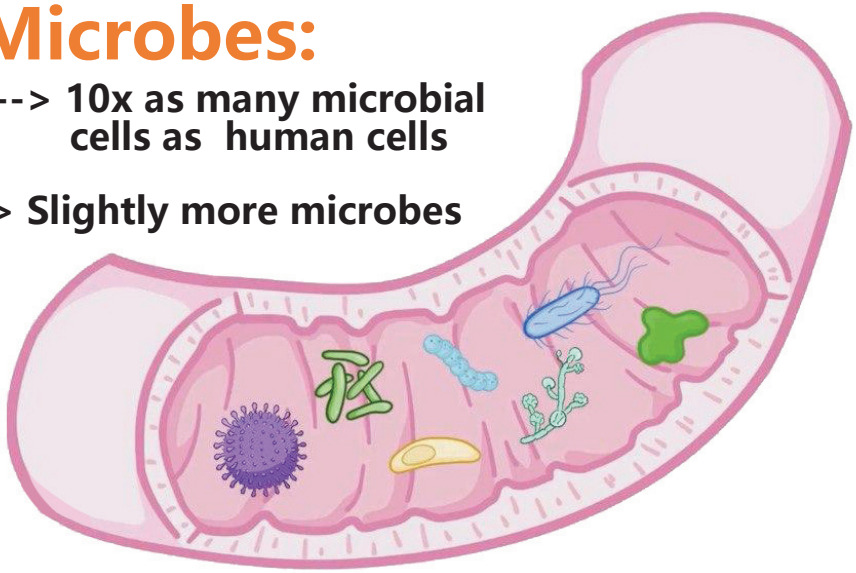
Trillions of Microbes:

- **Previously thought ---> 10x as many microbial cells as human cells**
- **Recent estimates ---> Slightly more microbes**



PROBIOTICS

- Live microorganism that offer a health benefit (eg. helping to enchance or restore health to our gut microbiome)
- Many microorganism that naturally live in our bodies (* Not all fermented foods are "probiotic") ---> Similiar to microorganism in probiotic foods,drinks & dietary supplements

* To be considered "probiotic" :
- sufficient living bacteria that survive food processing
- bacteria that survive have to benefit human health

- Lactobacillus (Firmicutes)
- Bifidobacterium (Actinobacteria)

PREBIOTICS

- Food components used by host microbes ---> offer health benefit

* Many prebiotics (* food for gut microbes) ---> High fiber foods that aren't broken down by human digestive enzymes ---> fermented by gut bacteria

- Inulin & Oligosaccharides ---> Prebiotics
- Most people don't eat enough fiber
- Average: 16g fiber/day
- Recommendation: 25-38g fiber/day

* Best sources of prebiotic fiber

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Digestive System

1. Gastrointestinal tract = 2nd brain

2. Gut microbiome (trillions of diverse organism including bacteria,-fungi, viruses) weighing between 2 to 6 pounds.
   - Break down dietary fiber
   - Support immune system
   - Helps keep out foreign invaders
   - Manufactures several essential vitamins

3. Prebiotics = high fiber foods (fruits, vegetables, beans and whole grains)
   - Microbes ferment the dietary fiber => fuel for the cells that line our intestinal tract
   - Whole foods : Bananas, onions, garlic, leeks, asparagus, arti chokes, tomatoes, plums, apples, nuts, soybeans and whole-wheat foods.
   - Functional food : Inulin, FOS, GOS (stimulate the growth of intestinal BIfidobacteria)

4. Probiotics = good bacteria => promotes gut health
   - Found in many fermented foods (covert sugars and other carbo hydrates in foods into acid)
   - Common source of probiotics (fermented dairy products such as yogurt, kefir and certain aged cheese => Lactobacillus acidophilus)
   - Fermented vegetables (pickles, kimchi, olives and sauerkraut)
   - Fermented soy (miso and tempeh)

5. Health benefits of Pre & Probiotics
   - Improves digestive health
   - Relieves constipation
   - Promotes weight loss
   - Helps control diabetes
   - Improved mineral absorption and bone health
   - May support heart health, prevent colon cancer and treat IBD

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